GTS123 Introduction to Computers and Programming

Lab 10: Problem 3

Write a Python program that loads students' raw grades from `grades.tsv`, calculates the GPA of all students, and displays the result. Part of the `grades.tsv` file is shown below:

```
6322882296 2.5 2.5 3.5 3.5 2.5 2.5
6322881601 2.0 3.0 2.5 2.5 2.5 3.5
6322882611 2.0 2.0 2.5 3.5 2.0 4.0
6322881065 2.5 3.5 2.0 4.0 3.0 2.0
...
```

From the data in the file, each student enrolls in 6 courses. The credits of the courses are 3, 2, 1, 3, 3, and 3, respectively. Therefore, the GPA of the first student (ID=6322882296) can be calculated as

$$\frac{(2.5 \times 3) + (2.5 \times 2) + (3.5 \times 1) + (3.5 \times 3) + (2.5 \times 3) + (2.5 \times 3)}{3 + 2 + 1 + 3 + 3 + 3} = 2.77$$

The output should look similar to the following.

```
Student ID      GPA
6322882296      2.77
6322881601      2.67
...
```

Note: To pass the test, your program must produce the exact output as shown above, including spaces and line breaks.